How-To Library

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Sizing Guidelines and Performance Tuning for Metadata Manager 10.0

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Abstract

This article is intended to provide you general reference guidelines to help you tune the performance of Metadata Manager. The results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of Metadata Manager might vary based on individual environments and other parameters even when using the same data.

The article lists the minimum system requirements and performance tuning guidelines for Metadata Manager 10.0 and is intended to be used by Informatica Global Customer Support and Professional Services.

Note: This article only includes database sizing information based on a PowerCenter resource. The article does not include sizing information for any other Metadata Manager resource type.

Supported Versions

- Metadata Manager 10.0.0

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Overview

Metadata Manager performance depends on the system environment where the Metadata Manager Service and associated databases run and the configuration of the Metadata Manager Service. This article describes the factors that affect performance, including system requirements, environment setup, and performance tuning options.

Note: This document lists the performance tuning and sizing guidelines for Metadata Manager installed in a split domain mode.

The following table lists the Metadata Manager operations and the factors that affect the performance of each operation:

Operation	Feature Type and Description
Resource load	- Environment - RAM and CPU - External - Size of source repositories - Network speed - Number of concurrent users - Tuning - Tablespace size - Number of concurrent resource loaded - Metadata Manager maximum heap size - Metadata Manager Agent memory, logging level, and validation level
Search indexing	- Environment - RAM, CPU, disk space - External - Size of source repositories - Network speed - Number of concurrent users
Searching	- Environment - RAM and CPU - External - Number of concurrent users - Disk I/O - Tuning - Search index optimization
Data lineage diagram	- Environment - RAM, CPU, and disk space - External - Number of concurrent users - Disk I/O - Tuning - Metadata Manager heap size - Lineage store size

Environment Setup

This article includes test data from internal testing of Metadata Manager performance. The system recommendations are based on the results of this testing. The test environment used to generate the results in this article includes two Red Hat Enterprise Linux servers. The machines were used to run a PowerCenter instance with Metadata Manager and the associated databases.

The following table lists the system resources for the two Linux servers, where Machine 1 hosts the database instance and Machine 2 runs the Metadata Manager Service:

Option	Machine 1 (64 bit)	Machine 2 (64 bit)
CPU	72 Core Intel(R) Xeon(R) CPU E5-2699 v3	72 Core Intel(R) Xeon(R) CPU E5-2699 v3
Clock speed	2.30 GHz	2.30 GHz
RAM	64 GB	64 GB

Option	Machine 1 (64 bit)	Machine 2 (64 bit)
Operating system version	6.6	6.6
Software and version	Oracle 11g	PowerCenter 10.0
Network	1 GB/s Ethernet	1 GB/s Ethernet

Note: No other operations were performed on the test machines during the Metadata Manager testing.

Metadata Manager Service System Requirements

The following table lists the minimum and recommended system requirements for a Metadata Manager instance as a result of internal testing:

Requirement	Minimum	Recommended
CPU or core	2	Same as the minimum requirement listed.
RAM	4 GB	8 GB
Metadata Manager warehouse tablespace	Up to 5x the size of the metadata in the source system (for PowerCenter).	Same as the minimum requirement listed.
Database temporary tablespace	At least 5 GB. You may need more depending on the number of resources and the type of resources that you load into the Metadata Manager warehouse. **Note:** For Cloudera Navigator resources, you can set a minimum of 2x the minimum database temporary tablespace size specified.	At least 10 GB. You might need more depending on the number of resources and the type of resources that you load into the Metadata Manager warehouse. **Note:** For Cloudera Navigator resources, the recommendation is to set 2x the amount of the recommended database temporary tablespace specified.
Search index file size	2 – 3 times the size of the metadata in the source system (for PowerCenter).	Same as the minimum requirement listed.
Processor	64 bit	Same as the minimum requirement listed.
Disk space	4 GB for one million lineage objects in the Metadata Manager warehouse.	Depends on the number of lineage objects in the Metadata Manager warehouse. For one million of lineage objects, the lineage graph occupies approximately 4 GB of disk space.
Disk I/O	140 MBps	-

One machine is the minimum required to run Metadata Manager. If you have higher number of concurrent users who perform different operations within Metadata Manager including resource load, use one machine to run the PowerCenter domain and the Metadata Manager application. Use another machine to host the database for the Metadata Manager warehouse and PowerCenter source repositories.

Metadata Manager Warehouse Tuning Parameters

The following table lists the tuning parameters along with the values configured for Metadata Manager warehouse on Oracle 11g:

Parameter Name	Value
db_block_size	8192
memory_target	10 GB. You can set this value up to 80% of the available RAM on the machine that hosts the database.
open_cursors	300
plsql_warnings	DISABLE:ALL
processes	500
sessions	792
SGA	4 GB. You can set this value up to 80 percent of the value specified for the memory_target parameter.
PGA	1 GB. You can set this value up to 20 percent of the value specified for the memory_target parameter.

Metadata Manager Max Heap Size

The Max Heap Size property for the Metadata Manager Service indicates the amount of RAM in megabytes allocated to the Java Virtual Machine (JVM) that runs Metadata Manager. Use this property to increase the performance of Metadata Manager. Default is 4 GB.

Set the Max Heap Size property for the Metadata Manager Service in the Administrator tool (Informatica Administrator).

The maximum heap size also affects CPU utilization. With a lower amount of memory and a higher number of concurrent users, the Metadata Manager Service process becomes single-threaded and uses only a single CPU. When Metadata Manager operates as a single-threaded application, all requests to the application become serialized. Additional requests must wait until the previous requests are processed.

The maximum heap size required increases as more concurrent users place a greater demand on the memory. A lower maximum heap size decreases Metadata Manager performance as the workload on the machine running the Metadata Manager Service increases. Both memory and disk I/O can affect performance. In general, the larger the maximum heap size, the greater the performance.

Note: You can clear the system level cache if you experience a lag in the performance of Metadata Manager. This might improve the performance of Metadata Manager.

Resource Load

The performance of Metadata Manager depends on the number of available CPUs and maximum heap size during resource load operations.

In general, the performance of Metadata Manager with a larger number of concurrent users increases as the number of CPUs and maximum heap size increases. The effect on load times does not include other factors, including searching and browsing operations during the metadata load.

Metadata Manager is a multithreaded application and can use up to 100% of CPU resources. If usage is at 100%, additional operations are serialized, which degrades Metadata Manager performance. As the number of concurrent users increases, additional CPUs are required to increase Metadata Manager performance.

Other factors that affect resource load performance include the speed of the network where Metadata Manager runs and the configuration of the source repositories. Slow network speeds or slow metadata source repositories also have a negative affect on load performance.

Tuning Resource Load Performance

The Max Concurrent Resource Load property for the Metadata Manager Services indicates the maximum number of resources that Metadata Manager can load simultaneously. If a resource load fails but can be resumed, Metadata Manager keeps the resource load in the queue until the timeout interval is exceeded. Default is 3. The maximum value for this property is 10.

Set the Max Concurrent Resource Load property for the Metadata Manager Service in the Administrator tool. For more information, see the *Informatica Application Service Guide*.

Metadata Manager Agent

The Metadata Manager Agent is a lightweight program that Metadata Manager uses to access metadata sources and extract source metadata. The metadata sources include source files or source repositories for certain business intelligence and data modeling tools.

If you load metadata from Business Objects, Cognos Impromptu, or Microstrategy, load metadata from metadata source files, or the Metadata Manager Service runs on a UNIX machine, you install and run the Metadata Manager Agent to load metadata. You install the Metadata Manager Agent as a service on a Windows machine.

You can configure the following Metadata Manager Agent properties:

- Java Virtual Machine memory
- Logging level
- Validation level

Increase Java Virtual Machine Memory

If you want to load a large amount of metadata for a resource type that uses the Metadata Manager Agent, you can increase the Java Virtual Machine (JVM) heap size to increase performance. For example, increase the JVM heap size to 1 GB if you load a large universe from Business Objects and receive the following error: `Not enough storage is available to process this command.`

Note: Maximum value for this parameter is 1 GB.

To increase the JVM heap size, perform the following steps:

1. Open MIRSetup.xml from the following location in a text editor: `<Metadata Manager Agent installation directory>/conf`
2. Add the following option to the <RunTime> section:`<Options>-Xmx1024m</Options>`
3. Save MIRSetup.xml.
4. Restart the Metadata Manager Agent.

Logging Level

During the metadata extraction process, Metadata Manager uses the Metadata Manager Agent to connect to metadata sources and extract metadata. Metadata Manager writes any log events generated during the metadata extraction process to the Metadata Manager Agent log file. The log events include calls to the Metadata Manager Agent from the Metadata Manager application and log events generated by the Metadata Manager Agent.

Metadata Manager writes to mm_agent.log in the following directory: `<INFA_HOME>/logs/<node name>/services/MetadataManagerService/<Metadata Manager service name>/`

You can modify the logging level to change the type of events included in mm_agent.log. You can configure one of the following logging levels:

- FATAL
- ERROR
- WARNING
- INFO
- DEBUG

To modify the logging level, perform the following steps:

1. Stop the Metadata Manager Agent.
2. Open log4j.xml in a text editor. log4j.xml is located in the following directory: `<INFA_HOME>/services/MetadataManagerService/mmapps/mm/WEB-INF/classes/log4j.xml`
3. Edit the value attribute for the priority element under the com.informatica.metadata.services.imas.impl.task.miti category.
   The following example shows the logging level set to DEBUG:

```
<category name="com.informatica.metadata.services.imas.impl.task.miti">
<priority value="DEBUG" />
<appender-ref ref="FILE_MMAGENT"/>
</category>
```

4. Save log4j.xml.
5. Restart the Metadata Manager Agent.

Search Indexing

Search indexing is part of the load operation. Search indexing is affected by the maximum heap size for Metadata Manager Service and the number of CPUs for the machine running Metadata Manager.

The size of index files generated for a specific resource in depends on the size of that resource. Index files are stored in the following directory: `<INFA_HOME>/services/MetadataManagerService/mm_files/MM/mm_index`
The size of the temporary database tablespace can also affect Metadata Manager performance when indexing a resource during the load process. The indexing operation can fail if the temporary database tablespace is not large enough. For example, For example, Metadata Manager requires approximately 11 GB of temporary tablespace to index a PowerCenter resource for an 800 MB PowerCenter repository.

Catalog Search

The time to search the metadata catalog increases as the number of users increases. However, for a higher number of users, increasing the maximum heap size slows the performance. Response time is similar with catalog search at 2 GB and 4 GB maximum heap sizes. However, 4 GB is recommended when performing additional operations, such as resource load and data lineage analysis, in addition to searching the catalog.

Searching the metadata catalog is also affected by the amount of metadata for the resources in the catalog and the number of resources in the Metadata Manager warehouse.

Tuning Catalog Search Performance

You can increase search performance by optimizing the search index in Metadata Manager. Metadata Manager incrementally updates an index for a resource when you make a change to the resource. The method that Metadata Manager uses to index the warehouse can create multiple index files for a single resource. As a result, searching through multiple files decreases search performance.

When you optimize the index, Metadata Manager creates a single index file for each resource, which increases search performance.

To optimize the search index, perform the following steps:

1. On the **Load** tab, click **Actions** > **Configure Search Index**.
   The **Configure Search Index** window appears.
2. Click **Optimize Entire Index**.
   Metadata Manager optimizes the index for each resource.

Data Lineage

The time to create and display a data lineage diagram is linearly proportional to the number of features or structures with connections and the number of displayed elements. Therefore, a larger number of elements with connections in the diagram results in a longer construction time for the lineage diagram.
The following system factors also affect data lineage performance:

- CPU. The time to perform data lineage analysis increases as the number of concurrent users performing data lineage analysis increases. Additional CPUs improve the time required to perform the data lineage analysis for a larger number of concurrent users
  During data lineage analysis, CPU usage decreases with a higher number of CPUs. For a larger number of concurrent users, CPU usage reaches 100% for all CPU configurations. Add CPUs to support a higher number of concurrent users.
- Maximum heap size. Data lineage performance increases as the maximum heap size for the Metadata Manager Service increases. Performing data lineage analysis, especially with a large number features or structures with connections or with a large number of display elements, is a memory-intensive operation. The time to perform data lineage analysis also increases as the number of users increases. However, for a higher number of users, a larger maximum heap size slows the rate of increase.
- Disk I/O: An increase in the disk I/O speed improves the data lineage performance.

Data Lineage Process

When you start data lineage analysis for an object in the Metadata Manager warehouse, Metadata Manager completes the following steps:

- Constructs the data lineage diagram. Metadata Manager analyzes the starting object and finds all the connected objects. Metadata Manager also constructs the overall diagram, expands the diagram to display the starting object, and converts the diagram to an XML file.
- Displays the data lineage diagram. Metadata Manager uses the XML file to display the data lineage diagram in the browser.

You can also get more information about the individual tasks to construct and render the data lineage diagram and the time to complete the tasks in mm.log.

Analysis Results

You can use this section as a reference to compare your requirements and tune your Metadata Manager sizing and performance parameters. The values listed in the following sections are results of different analysis tests done at Informatica.

Lineage Graph Size Analysis

You can see the following table to tune your lineage store size:

Resource Type and Approximate Number of Entities	Approximate Number of Lineage Links (in Million)	Approximate Number of Expressions (in Million)	Approximate Lineage Graph Size (in GB)
- Data integration resources with 3200 mappings. - Database resources with 28000 tables. - 14000 reports	1.5	0.4	5
- Data integration resources with 5000 mappings. - Database resources with 20000 tables. - 60000 reports	2.3	0.6	7
- Data integration resources with 4500 mappings. - Database resources with 2000 tables. - 28000 reports	3.6	0.5	11

Note: As listed in the table, an increase in the number of lineage links between objects in the resources increases the lineage graph size.

Resource Load Analysis

You can see the following tables to determine the Metadata Manager warehouse size for the loaded resources and the entities included in the resources:

Data Integration Resources

Approximate Number of Mappings	Approximate Number of Objects	Approximate Metadata Manager Warehouse Size (in GB)
1000	22000	0.4
1300	38000	1
2100	56000	2.5

Database Resources

Approximate Number of Tables	Approximate Number of Objects	Approximate Metadata Manager Warehouse Size (in GB)
1000	18000	0.02
1400	54000	0.05
1700	86000	0.2

Cloudera Navigator Resource

Approximate Number of Hive Tables	Approximate Lineage Graph Size (in GB)	Approximate Metadata Manager Warehouse Size (in GB)
35000	7	64

Data Modelling Resources

Approximate Number of Tables	Approximate Number of Objects	Approximate Metadata Manager Warehouse Size (in GB)
600	21000	0.06
700	37000	0.2

Business Intelligence and Analytics Report Resources

Approximate Number of Reports	Approximate Number of Objects	Approximate Metadata Manager Warehouse Size (in GB)
200	7500	0.01
500	17500	0.05

Oracle Business Intelligence Enterprise Edition Resources

Approximate Number of Reports	Approximate Number of Objects	Approximate Metadata Manager Warehouse Size (in GB)
4600*	405000	0.14

*The approximate size of a report was 30 KB.

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